STAT 530 Bioinformatics				
Day	**Date**	**Topic**	**Lecture**	**HW due**
T	01/21/25	Introduction	Course overview	
Th	01/23/25		Principles	
T	01/28/25	Biology	Concepts	
Th	01/30/25		Technologies	
T	02/04/25		Paper	1
Th	02/06/25	Statistics	Concepts	
T	02/11/25		Covariation	2
Th	02/13/25		Inference	
T	02/18/25	Programming	Concepts	3
Th	02/20/25		VSCode	
T	02/25/25	Pseudobulk analyses	Data and preprocessing	4
Th	02/27/25		Formulate questions	
T	03/04/25		In-class work	5
Th	03/06/25		In-class work	
T	03/11/25		In-class work	6
Th	03/13/25		Challenge problem	
T	03/18/25		--- *SPRING BREAK* ---	
Th	03/20/25		--- *SPRING BREAK* ---	
T	03/25/25	Subregion analyses	Formulate questions	7
Th	03/27/25		In-class work	
T	04/01/25		Challenge problem	8
Th	04/03/25		In-class work	
T	04/08/25	snRNA-seq analyses	Preprocessing	9
Th	04/10/25		Formulate questions	
T	04/15/25		In-class work	10
Th	04/17/25		In-class work	
T	04/22/25		Challenge problem	11
Th	04/24/25		In-class work	
T	04/29/25	New analyses	Formulate questions	12
Th	05/01/25		In-class work	
T	05/06/25	Review	Review	13
T	05/13/25		**Final project due**	

STAT 530 Bioinformatics

Day	Date	Topic	Lecture	HW due
T	01/21/25	introduction	overview of field (not a science and not a tech but r	
Th	01/23/25		What is bioinf (meaning and understanding, linl	
T	01/28/25	biology	basics	1
Th	01/30/25		technologies and preprocessing (mention others esp	
T	02/04/25		paper, refer back to technologies	2
Th	02/06/25	statistics	population, features (latent is important), sample, v	
T	02/11/25		covariaiton/Decomposition, confoundi	3
Th	02/13/25		modeling, estimation, testing, multiple testing,	
T	02/18/25	coding	basics of programming, python, and r,	4
Th	02/20/25		vscode and ai	
T	02/25/25	spatial pseudobulk an	preprocessing (qc + norm), get data	5
Th	02/27/25		Formulate preprocessing and paper questions	
T	03/04/25		software packages and in-class analysis	6
Th	03/06/25		In-class analysis	
T	03/11/25		In-class analysis	7
Th	03/13/25		challenge problem (grn)	
T	03/18/25		--- *SPRING BREAK* ---	
Th	03/20/25		--- *SPRING BREAK* ---	
T	03/25/25	subregion analysis	Formulate paper questions	8
Th	03/27/25		In-class analysis (basically same as before tho may r	
T	04/01/25		challenge problem (cluster should b	9
Th	04/03/25		In-class analysis	
T	04/08/25	Snrna-seq analysis	preprocessing (qc + norm), get data	10
Th	04/10/25		Formulate paper questions	
T	04/15/25		software packages and in-class analysis	11
Th	04/17/25		In-class analysis	
T	04/22/25		In-class analysis	12
Th	04/24/25		challenge problem (?)	
T	04/29/25	challenge	imputation	13
Th	05/01/25		crosstalk across regions	
T	05/06/25		review, place everything we did into in	14

final project

are oligodendrocytes really there (deconv, impo
how much is location reflected in expression?
density estimation? Knns?

ight in the middle, give michael example, practice not theory) and structure of class
king quant and subst)
set up machine
p atac)
sequencing and alignment, multiplexing, coding theory
ariation
read paper and divide into questions
overfitting, training/test
practice framing stats questions

coding practice

do analysis in r

do analysis in python

bulk challenge problems, write your own and then use existing
need glmm)
r and python

Subregion challenge problem

r

python

scrnaseq challenge, write own and use existing

answer both challenge questions

ortant for spatial)